

Electronic Circuit Design (EE2001)

Date: November 5th 2024

Course Instructor

Dr. Omer Saleem

Sessional-II Exam

Total Time (Hrs): 1

Total Marks: 30

Total Questions: 3

22L-6234

Roll No

3A

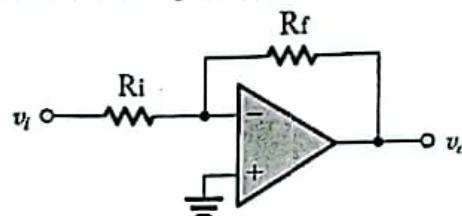
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CLO # 3: Appraise the limitations of real op-amp circuits.

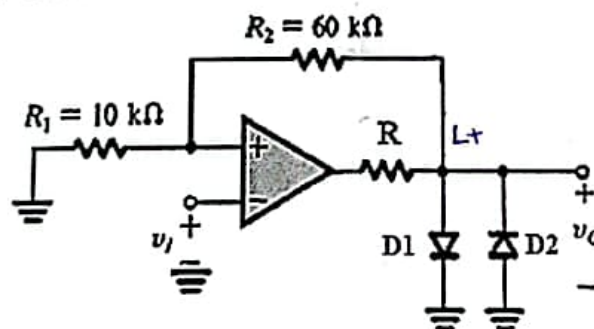
Q1: An imperfect op-amp is connected as shown below, with a $R_i = 100 \text{ k}\Omega$ and $R_f = 10.0 \text{ M}\Omega$. The output voltage is found to be +9.35 V when a non-zero DC input voltage v_i is applied, and +7.10 V when the input is grounded (i.e. $v_i = 0$). Find the magnitude of v_i and the offset voltage V_{offset} . [10 marks]



CLO # 4: Design wave-form generation circuits.

Q2: Refer to the following bistable multivibrator circuit. [10 marks]

- Sketch and label the transfer characteristics of v_o - v_i . The diodes are assumed to have a constant 0.7 V drop when conducting, and the immediate output of the op-amp saturates at $\pm 12 \text{ V}$.
- Find the value of R , such that the maximum diode current in $D1$ is 1.12 mA when the op-amp output is saturated at +12 V.



CLO # 4: Design wave-form generation circuits.

Q3: Design a Wien-Bridge oscillator that generates a sine wave of frequency 9.95 kHz. [10 marks]

- Draw the circuit schematic.
- Find the values of all circuit components, using a minimum resistance of 150 kΩ in the circuit.